MIL-D-50028A(MU)
6 July 1970
SUPERSEDING
MIL-D-50028(CmlC)
24 June 1959

MILITARY SPECIFICATION

DB3

1. SCOPE

1.1 This specification covers one grade of DB3 which is chemically known as 4-(4-nitrobenzyl) pyridine.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-676 - Boxes, Setup.
PPP-C-186 - Containers, Packaging and Packing For Drugs, Chemicals, and Pharmaceuticals.
PPP-T-45 - Tape, Gummed, Paper, Reinforced and Plain, For Sealing and Securing.

MILITARY

MIL-P-116 - Preservation, Methods Of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-650 - Explosive: Sampling, Inspection and Testing.

FSC 6810

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNFIORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Freight Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the National Classification Board, 1616 P Street, N.W., Washington, D.C. 20036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D1193-66 - Reagent Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Melting characteristics. DB3 shall begin to melt at a temperature of no less than 68° C and shall be completely melted at a temperature of no more than 72° C when tested as specified in 4.2.4.1.

3.2 Acetone solubility. DB3 shall produce no turbidity in acetone when tested as specified in 4.2.4.2.

3.3 Benzyl chloride reaction. DB3 shall develop a color density of more than 90 Klett units when tested as specified in 4.2.4.3.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the DE3 produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of preparation for delivery. Sampling for examination of preparation for delivery shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test. Sampling for test shall be conducted in accordance with table I. A representative specimen of approximately 10 grams (g) shall be removed from each sample unit packaging container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

Table I. Sampling for test

Number of unit packaging containers in batch or lot	Number of sample unit packaging containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

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4.2.3 Inspection procedure.

4.2.3.1 For examination of preparation for delivery. The sample unit shall be one filled unit, intermediate, or shipping container, as applicable, ready for shipment. Sample containers and the preparation for delivery thereof shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Slotted partitions missing or not as specified (where required)
(f) Marking incorrect, missing, or illegible

4.2.3.2 For test. Approximately equal portions of all of the specimens taken in 4.2.2.2 shall be thoroughly mixed to form a composited specimen of no less than 20 g. The composited specimen shall be tested as specified in 4.2.4. Each test shall be conducted in duplicate analysis. Failure of either analysis of any test shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Melting characteristics. Determine the temperature at which the specimen begins to melt and the temperature at which the specimen is completely melted in accordance with method 209.1 of MIL-STD-650 except do not dry the specimen prior to testing. If applicable, use the calculation for temperature correction specified in method 209.1.

4.2.4.2 Acetone solubility. Weigh 1.0 ± 0.1 g of the specimen into a 10-milliliter (ml) test tube and add 5 ml of freshly distilled acetone. Agitate until the specimen dissolves and observe for turbidity against a white background.

4.2.4.3 Benzyl chloride reaction.

(a) Specimen solution. Weigh 1.05 ± 0.01 g of the specimen into a 25-ml volumetric flask. Add 10 ml of n-propanol, swirl to dissolve the specimen, add 5 ml of water, and mix. Dilute to the 25-ml mark with n-propanol. Prepare a fresh solution daily and protect it from sunlight.

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(b) Benzyl chloride solution. Prepare a solution of benzyl chloride in n-propanol by pipeting 3.7 ml of benzyl chloride into a 25-ml volumetric flask and diluting to the mark with n-propanol (1 ml of this solution will contain 0.163 g of benzyl chloride at 20° C). Prepare a fresh solution daily and protect it from sunlight.

(c) Procedure. Pipet 0.10 ml of specimen solution prepared as specified in (a) into a test tube. Add 1 ml of benzyl chloride solution prepared as specified in (b) and 4.9 ml of n-propanol and mix. Immerse the test tube for 10 minutes ± 5 seconds in a constant temperature bath maintained at a temperature of 100° ± 1° C so that the contents of the test tube are at least 1 inch below the surface of the bath. Remove the test tube and immerse it for 5 minutes in a second constant temperature bath maintained at a temperature of 25° ± 3° C so that the contents of the test tube are at least 1 inch below the surface of the bath. Remove the test tube and quantitatively transfer its contents to a calibrated Klett colorimeter tube using small quantities of acetone. Dilute with acetone to the 10-ml mark on the calibrated tube. Add 1.0 ml of piperidine and mix thoroughly with a stirring rod. Exactly 2 minutes after adding the piperidine, read the optical density on a Klett-Summerson Photoelectric Colorimeter (which has been adjusted to read zero against a tube of water) using a Klett No. 54 green filter. Run a second determination, substituting 1 ml of n-propanol for the 1 ml of benzyl chloride solution and obtain a reading. Subtract the second reading from the reading obtained using the benzyl chloride solution to obtain the net Klett reading.

5. PREPARATION FOR DELIVERY

5.1 Packaging, level C.

5.1.1 Unit packaging. DB3 shall be packaged in unit quantities of 50 g, 100 g, or 1 pound as specified (see 6.2).

5.1.1.1 Fifty-gram quantity. A quantity of 50 ± 0.25 g of DB3 shall be packaged in a new, clean, widemouth glass vial conforming to group A, class 1, type e, style 6, grade 1, with outerseal A of PPP-C-186 in accordance with applicable requirements specified therein. The vial shall be closed by means of a ground glass stopper, formed from the same material as the vial, mated to the neck of the vial.

5.1.1.2 One hundred-gram quantity. A quantity of 100 ± 0.5 g of DB3 shall be packaged as specified for the 50-g quantity in 5.1.1.1 except that a widemouth glass bottle conforming to style 2 of PPP-C-186 shall be used instead of the style 6 glass vial. The bottle shall be closed by means of a ground glass stopper, formed from the same material as the bottle, mated to the neck of the bottle.

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5.1.1.3 One-pound quantity. A quantity of 1 ± 0.1 pound of DB3 shall be packaged as specified for the 100-g quantity in 5.1.1.2.

5.1.2 Intermediate packaging.

5.1.2.1 Fifty-gram quantity. Twenty glass vials of DB3, packaged as specified in 5.1.1.1, shall be intermediately packaged in accordance with method IC-3 of MIL-P-116 using either a variety 1, style IV, type optional, class i or j paperboard folding box of PPP-B-566; or a type I, variety 1, class and style optional paperboard set-up box of PPP-B-676. Each vial shall be placed in a close-fitting cell formed by the use of slotted partions within the box. The partitions shall be high enough to prevent the vials from touching each other and shall be made of the same material as the box. Each box shall be closed by 2-inch wide gummed tape conforming to type III, grade A of PPP-T-45 as specified in the appendix to the applicable box specification.

5.1.2.2 One hundred-gram quantity. Ten glass bottles of DB3, packaged as specified in 5.1.1.2, shall be intermediately packaged as specified for 50-g quantity in 5.1.2.1 except for the number of unit containers and the size of the box.

5.2 Packing, level C. Ten boxes, intermediately packaged as specified in 5.1.2.1 or 5.1.2.2; or 20 1-pound bottles, packaged as specified in 5.1.1.3, shall be packed in a fiberboard box which complies with the Uniform Freight Classification Rules, the National Motor Freight Classification Rules, or other carrier rules as may be applicable to the mode of transportation. The box used for 1-pound bottles shall have a cell for each bottle formed from slotted fiberboard partitions so that the bottles are separated from each other. All packing boxes shall be capable of being stacked and of supporting superimposed loads during shipment and shall assure safe delivery to destination without damage to contents. The packing boxes shall be acceptable for shipment at the most favorable rate of the applicable regulation provided that all requirements specified herein have been met.

5.3 Marking. Unit, intermediate, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. DB3 is intended for use in chemical agent detection kits to detect mustard agents.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Unit quantity required (see 5.1.1).

6.3 Batch. A batch is defined as that quantity of material has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated valve should be rounded off "tc the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E29).

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

Project No. 6810-0605

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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